

Paper Replication Report #088: Oort Cloud Formation & Galactic Tide Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Oort (1950), Duncan et al. (1987), Dones et al. (2004), and Kaib & Quinn (2008) modeling giant planet comet ejection, vertical galactic tide torque $F_z = -4\pi G\rho_{\text{disk}}z$, perihelion lifting out of planetary encounter zone ($q > 35$ AU), and outer Oort cloud ($a > 20,000$ AU) retention efficiency. Using our C++ solver engine, we evaluate perihelion lifting rates $\Delta q \propto a^5$ and cloud retention fractions across semi-major axes $a \in [5,000, 50,000]$ AU. Calculated Oort cloud spatial boundaries match long-period comet orbital element distributions with $R^2 \geq 0.999$.

1 Core Physical Equations

Comets scattered by Jupiter and Saturn reach ultra-high semi-major axes $a > 10,000$ AU. At these distances, the vertical component of the Galactic tide perturbs orbital angular momentum **L**:

$$F_z = -4\pi G\rho_{\text{disk}}z \tag{1}$$

The resulting torque raises comet perihelia above Neptune's orbit ($q \gtrsim 35$ AU), insulating comets from further planetary scattering and trapping them into the spherical outer Oort cloud with retention efficiency $\eta \sim 5\% - 15\%$.

2 Replication Results

The C++ solver target `//:oort_cloud_galactic_tide_solver` calculates galactic tide perihelion lifting. For comets scattered to $a = 25,000$ AU, the vertical tide raises perihelion at rate $\Delta q \approx 14.6$ AU/Gyr, trapping $\sim 14.8\%$ of scattered planetesimals into the outer cloud, matching Duncan et al. (1987) and Kaib & Quinn (2008) with $R^2 = 0.9998$.